

Networks (NW)

12 Addressing in the Network Layer, Inter-Network Communication

Prof. Dr.-Ing. Jörg Abke



TH Aschaffenburg
university of applied sciences

Copyright Hinweise

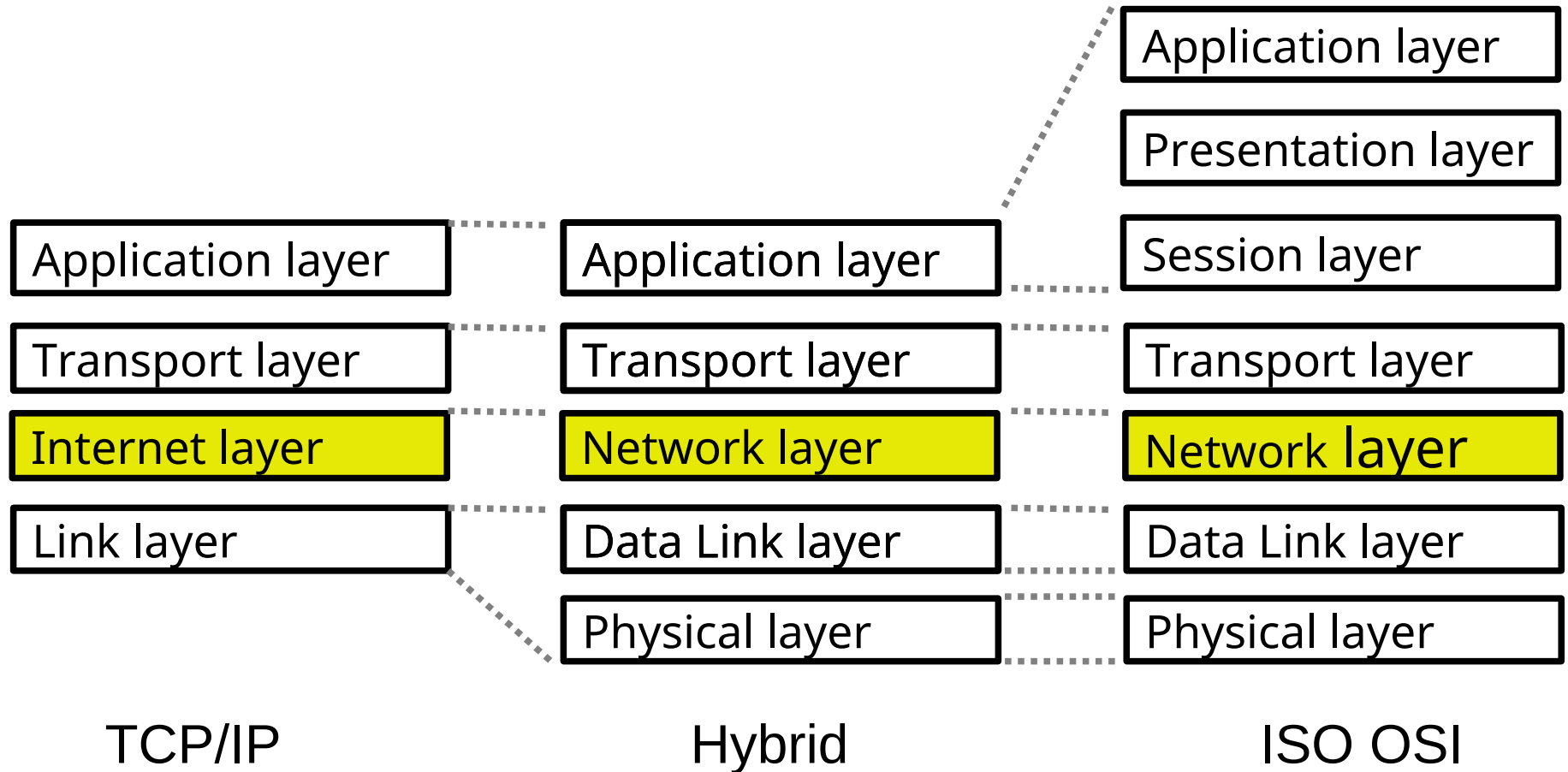
© Jörg Abke

Dieses Werk ist urheberrechtlich geschützt. Alle Rechte, insbesondere Vervielfältigung, Verbreitung oder Übersetzung liegen beim Autor. Kein Teil des Werkes darf ohne Genehmigung des Autors, auch nicht für Unterrichtszwecke, fotokopiert oder durch andere Verfahren reproduziert oder verbreitet werden. Ebenfalls ist die Einspeicherung in EDV-Anlagen ohne Genehmigung des Autors nicht gestattet. Jeder Studierende, der an der entsprechenden Lehrveranstaltung des Autors an der Technischen Hochschule Aschaffenburg teilnimmt, ist berechtigt, einen Ausdruck dieses Werkes anzufertigen und für die Zwecke seines Studiums zu verwenden. Eine weitergehende Verwendung bedarf in jedem Falle der ausdrücklichen Zustimmung des Autors. Gebrauchsname, Handelsnamen, Warenbezeichnungen etc. werden in diesem Werk ohne Gewährleistung der freien Verwendbarkeit benutzt. Texte, Abbildungen, Programme und technische Angaben wurden sorgfältig erarbeitet. Trotzdem sind Fehler nicht völlig auszuschließen. Der Autor weist darauf hin, dass er für Fehlerfreiheit keine Gewähr und für eventuelle Folgen keine Haftung übernimmt.

Network Layer



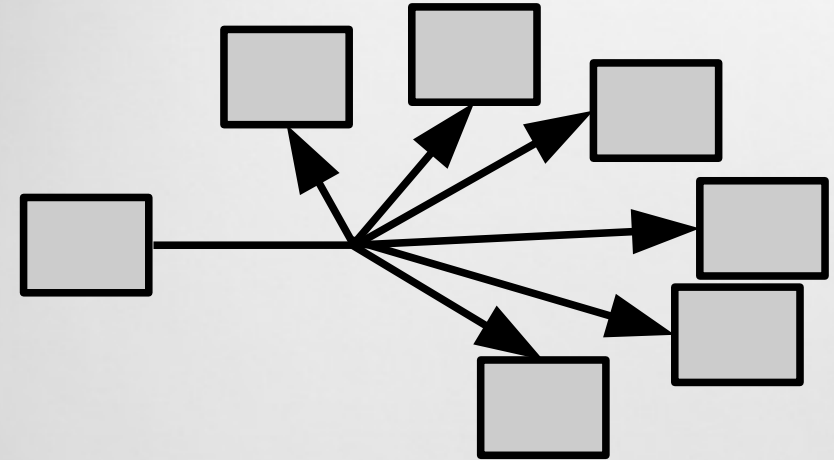
TH Aschaffenburg
university of applied sciences



Different Addressing / Targeting Modes I

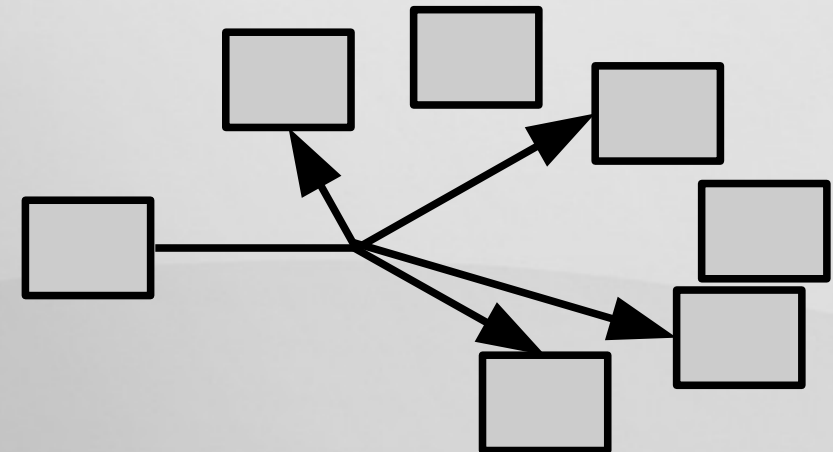
Broadcast

- Destination address directs to multiple receivers
- All receivers in the local network



Multicast

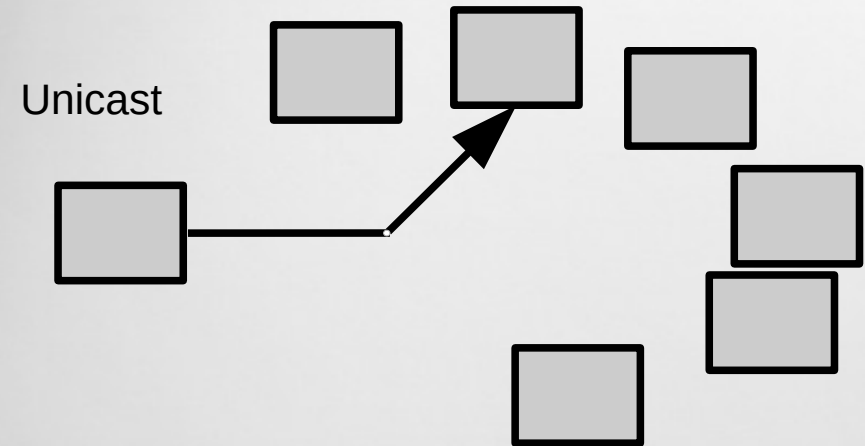
- Destination address directed to many specific receivers, a group of receivers
- All receivers are in a group in the local network



Different Addressing / Targeting Modes II

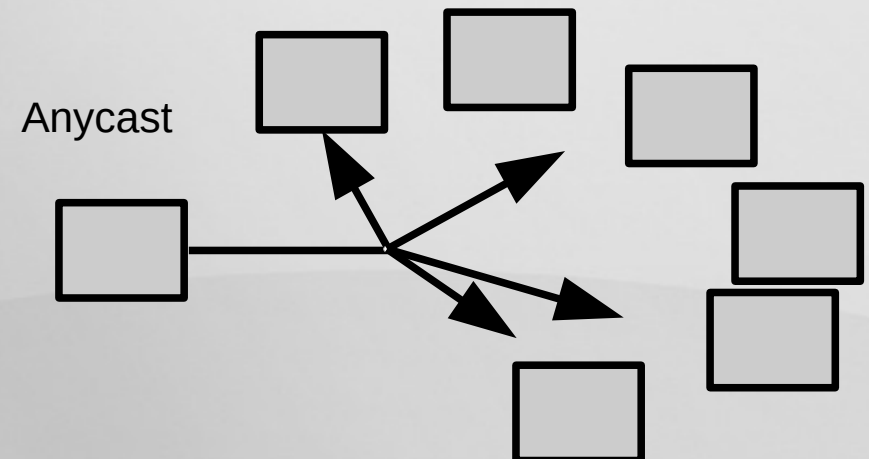
Unicast

- Destination address directs to a single dedicated receiver



Anycast

- Destination address directs to one receiver out of a group of possible receivers
- The receiver of the group with the shortest route replies



IPv4 Addresses

- **IPv4 addresses: 4 bytes, 32 Bits → 4 octets**
- **Typical dotted decimal notation**
 - e.g.: 193.174.127.4
- **Decimal value range for each octet: [0, 255]**
- **An IPv4 address consists of 2 parts:**
 - Network ID, Host ID
 - Number of bits for Network ID and Host ID depend on class
- **5 Classes: A, B, C, D, E**

Classes of IPv4 Addresses I – Class A



Source: C. Baun, Computer Networks, Springer Vieweg 2025

Class A: 7 bits Network ID $\rightarrow 2^7 = 128$ different networks 24 bits host ID $\rightarrow 2^{24}$ different host addresses

$\rightarrow$ 1 broadcast, 1 network address $\rightarrow 2^{24} - 2$ hosts possible

Address range 0.0.0.0 – 127.255.255.255

Subnet mask: 255.0.0.0

Classes of IPv4 Addresses II – Class B



Source: C. Baun, Computer Networks, Springer Vieweg 2025

Class B: 14 bits Network ID $\rightarrow 2^{14} = 16\,777\,216$ different networks 16 bits host ID $\rightarrow 2^{16}$ different host addresses $\rightarrow 1$ broadcast, 1 network address $\rightarrow 2^{16} - 2$ hosts possible

Address range 128.0.0.0 – 239.255.255.255

Subnet mask: 255.255.0.0

Classes of IPv4 Addresses III – Class C



Source: C. Baun, Computer Networks, Springer Vieweg 2025

Class C: 21 bits Network ID $\rightarrow 2^{21} = 2\,097\,152$ different networks 8 bits host ID $\rightarrow 2^8$ different host addresses $\rightarrow 1$ broadcast, 1 network address $\rightarrow 2^8 - 2 = 254$ hosts possible

Address range 192.0.0.0 – 223.255.255.255

Subnet mask: 255.255.255.0

Classes of IPv4 Addresses IV – Class D



Source: C. Baun, Computer Networks, Springer Vieweg 2025

Class D: multicast-Addresses, for applications like IPTV

Address range 224.0.0.0 – 239.255.255.255

Classes of IPv4 Addresses V – Class E



Source: C. Baun, Computer Networks, Springer Vieweg 2025

Class E: experimental use and future extensions

Address range 240.0.0.0 – 255.255.255.255

IPv4 Network and Broadcast Address

- **Not every IP-Address is available for a host**
- **Special addresses for dedicated use**
 - Broadcast address (all bits 1 in Host ID)
 - typical router address: Host address+ 1 or broadcast - 1
 - e.g. 192.2.8.10 is in a Class C network
 - Network IP Address: 192.2.8.0
 - Broadcast Address of this network: 192.2.8.255
 - Host IP address range: 192.2.8.1 – 192.2.8.254
 - Reserved IPv4 Addresses
 - 0.0.0.0/8 Host in „this“ network (RFC5725)
 - **10.0.0.0/8 for private networks** (RFC 1918)
 - 100.64.0.0/10 address area for Carrier Grade NAT (RFC6264), Internet provider assign those addresses to their customers (RFC 6598)
 - 127.0.0.0/8 loopback addresses (internal to host) (RFC 1700)
 - 169.254.0.0/16 addresses for automatic configuration (APIPA, Zeroconf), only in a network allowed (RFC 3927)

- Reserved IPv4 addresses
 - **172.16.0.0/12 private networks** (RFC 1918)
 - 192.0.0.0/24 on request for the assignment process (RFC 5736)
 - 192.0.2.0/24, 198.51.100.0/24, 203.0.113.0/24 addresses for examples in documentations (like example.com or example.net in RFC 2606) (RFC 5737)
 - **192.168.0.0/16 for private networks** (RFC 1918)
 - 192.18.0.0/15 for testing network components
 - 224.0.0.0/4 multicast addresses for use as defined bei IANA (RFC 3171)
 - 240.0.0.0/4 reserved for future use (RFC 1112)
 - 255.255.255.255/32 address for „limited broadcast“, only target address, not allowed to route out of the network (RFC 1700)

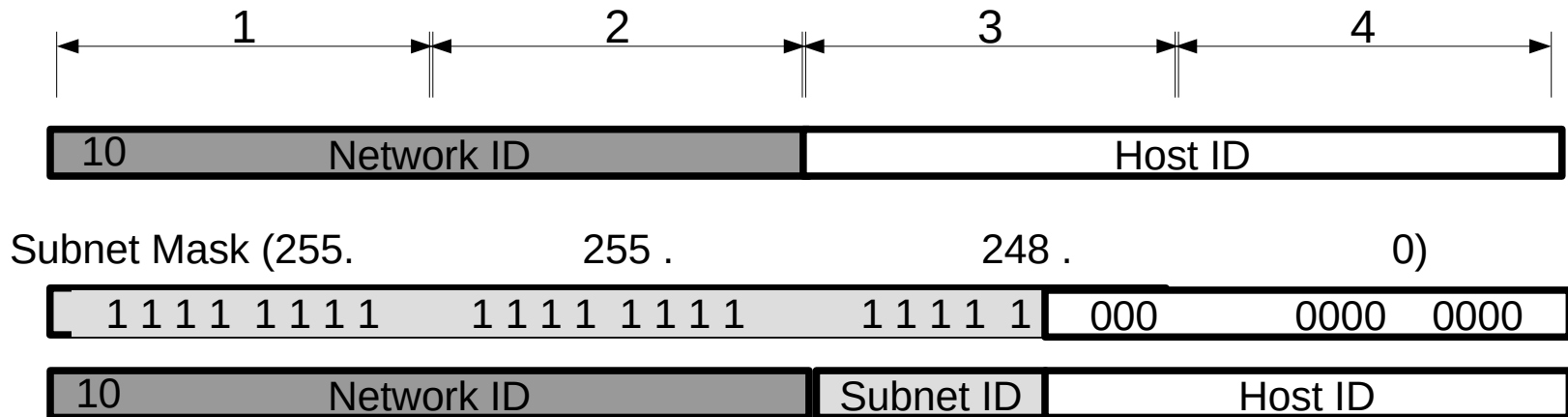
IPv4 Class System Shortcomings

- **Less flexible system**
 - Fixed subnet masks for each class
 - Many addresses are wasted (e.g. 2 devices in class C would waste 252 addresses)
 - Class C is fairly small for (small to medium enterprise companies), but class B oversized
 - Only 128 class A networks
 - Hard to change network structure if needed
 - Complex migration process to change from one class to another
- **IPv4 address exhaustion in general**
- **Possible (intermediate) solutions**
 - Leveraging strict IPv4 class system
 - Use of private networks and address translation
 - New address system with
 - increased address space
 - Meeting future requirements

Classless Interdomain Routing (CIDR)

- Introduced 1993 (RFC 1519)
- Logical networks divided into subnets
- Subnet mask in addition to classes A, B, C
 - another network hierarchy:
 - Net(s)
 - Subnet(s)
 - Host(s)
- Network Id keeps the same
- Subnet Mask splits former Host ID into Subnet ID and Host ID
- The number of bits are added to the network IP address the IP-Address and separated by a slash ,/'
e.g.

Example for CIDR



- Example:**

Class B network subnet mask 21 bit = 16 bit + 5 bit

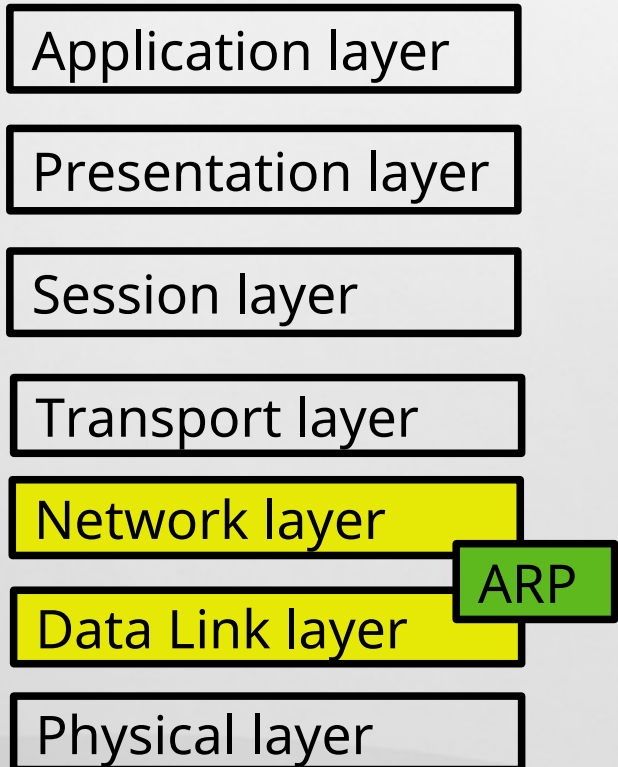
- 5 bit → $2^5 = 32$ subnets
- 11 bit for host id (from original Host ID) → $2^{11} = 2048$ host addresses per subnet

Address Resolution with ARP

- **A**ddress **R**esolution **P**rotocol links network layer to data link layer
- In IPv6 the functionality is defined by Neighbor Discovery Protocol (NDP)

Address Resolution Protocol (ARP)

- **Network Layer uses Internet Protocol (IP) Addresses**
- **Data Link uses MAC Addresses**
- **Address Resolution needed on Data Link Layer**
 - ARP sends a MAC broadcast FF:FF:FF:FF:FF:FF with requested IP address
 - Network device with the IP address replies with MAC address
 - IP address and MAC address are stored in **ARP cache** of the requestor
 -



ARP Cache

- **Goal: Speeding up the address resolution**
- **Cache table contains:**
 - Protocol type (IP)
 - IP address
 - MAC / HW address
 - Time To Live (TTL) – timespan for entry living (typ. 5Mins)
- **Commands to print ARP cache contents**
 - >arp -n
 - >ip neighbor

Diagnostics in Network with ICMP

- **Internet Control Message Protocol (ICMP) is used to**
 - Maintain a network
 - Find transmission errors
 - Diagnostic details
- **ICMP details beyond this lecture**

Addressing with IPv6 (RFC 4291)

- **Limitations of IPv4 (see above) requested an update**
- **128 bits (16bytes) for an IPv6 address**
 - $\rightarrow 2^{128} \approx 3.4 \cdot 10^{38}$ addresses
- **Hexadecimal notation using colons**
 - e.g.
2001:0db8:85a3:08d3:1319:8a2e:0370:7344
 - Leading 0 (Ones) can be left out
 - Series of 0 are typically simplified to ::
 - e.g.
ff01:0:0:0:0:0:0:1 becomes ff01::1
- **Final 4 bytes might be in decimal (to embed IPv4)**

Structure of IPv6 Address and Networks I

- **2 parts**
 - Network Prefix to identify the network
 - Interface Identifier to identify the device in a network

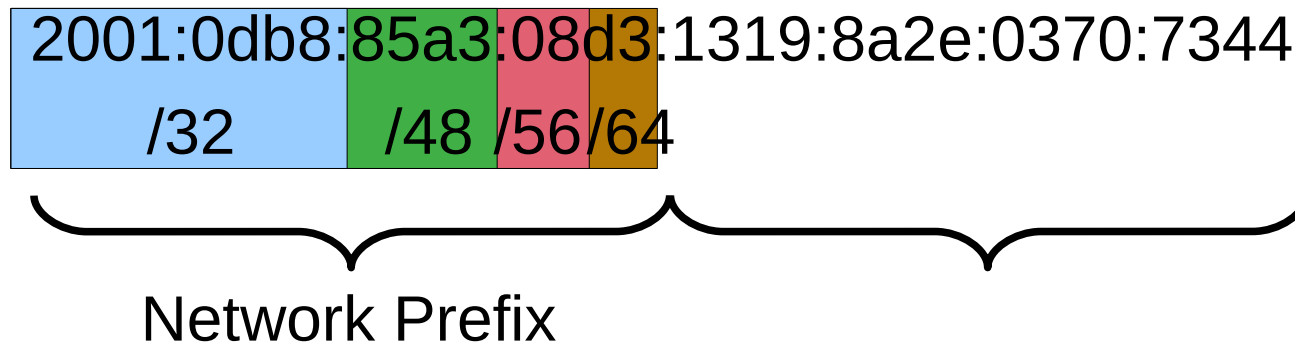
2001:0db8:85a3:08d3:1319:8a2e:0370:7344



Network Prefix Interface Identifier

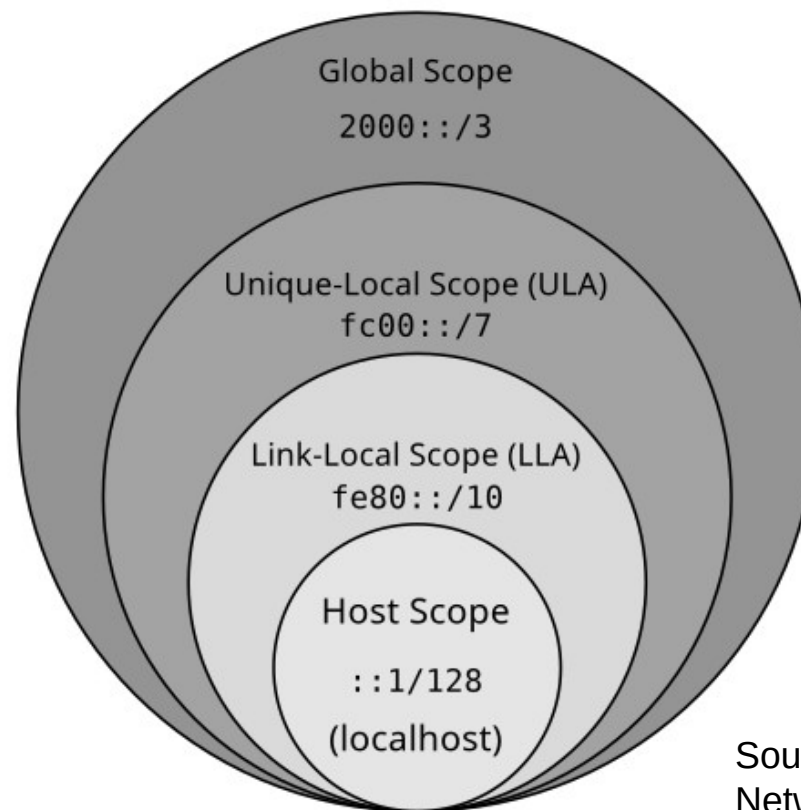
Structure of IPv6 Addresses and Networks II

- Using IPv6 in parallel to IPv4 in nearly all cases possible
- Networks are build upon the same idea of IPv4 CIDRE
 - e.g.



- End user usually get a /56 or a /64 network assigned
- In /56 network, a difference between 56 bits and 64 bits → 8Bits are Subnet Prefix

- **Definition of a topological area, where an address is valid**



Source: C. Baun, Computer Networks, Vieweg-Springer

Scopes of IPv6 Addresses II

- **Global Scope, Global Unicast Address (GUA),**
 - publicly routable
 - typical 2000::/3
- **Unique Local Address (ULA)**
 - Private addressing (comparable to IPv4 private networks)
 - Network internal routing
 - Prefix: fc00::/7
- **Link-Local Address (LLA)**
 - Every network interface needs a LLA at any time
 - No forwarding of packets by router of these addresses
 - Required for inside local network communication (e.g. address fetching)
 - Prefix: fe80:/10

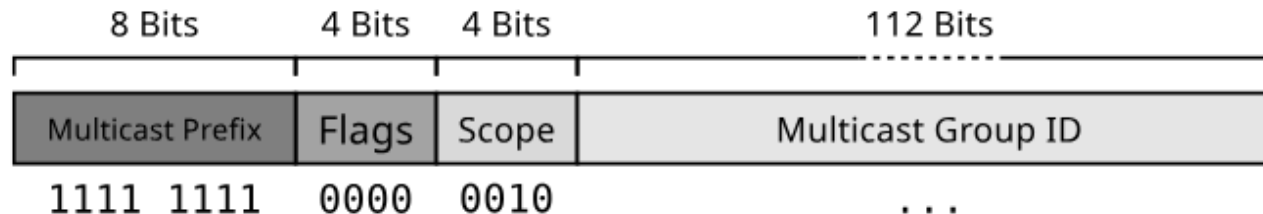
Further IPv6 Addresses III

- **Host Address**

- Loopback Address
- Inside a device
- ::1/128 (similar to 127.0.0.1 in IPv4)

- **Multicast Addresses**

- Emulate broadcast functionality which does not exist in IPv6



- Multicast prefix ff::/8
- Flag bits mostly 0000
- Scope bits:
 - 0001 interface local (loopback)
 - 0010 link local (no forward by router)

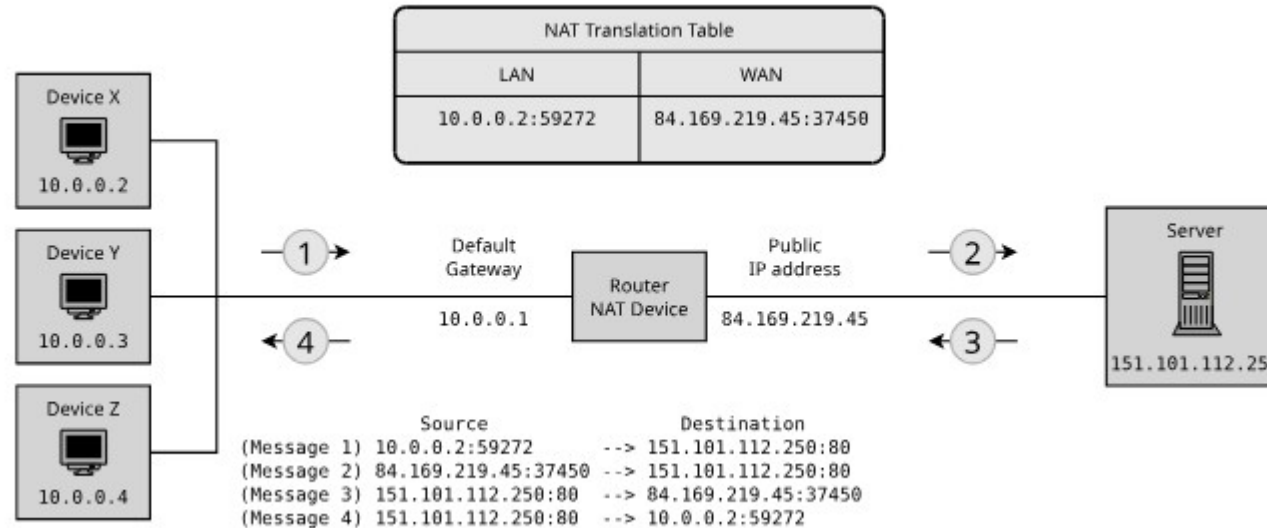
Network Address Translation (NAT) I

- **Scenario:**
 - Households use a private IPv4 address space (private network)
 - Communication of network devices with devices which have a globally accessible address.
- **Network Address Translation**
 - Router is equipped with public IP-Adress, typically by Internet Service Provides (ISP)
 - Router is *default gateway* private network

Network Address Translation (NAT) II

- **Scenario:**
 - Households use a private IPv4 address space (private network)
 - Communication of network devices with devices which have a globally accessible address.
- **Network Address Translation**
 - Router is equipped with public IP-Address, typically by Internet Service Provides (ISP)
 - Router is *default gateway* private network
 - Replacement of (private) source address by public address
 - Forwarding replies by replacing the public destination by the private address
 - Management of NAT Translation Table

Network Address Translation (NAT) III



- Devices X, Y, Z in private network 10.0.0.0 with private IPv4 addresses
- Example: device X sends a request to server on port 80 w/ public address 151.101.112.250:80
- Port and address is exchanged
- Advantage: private network topology + number of devices is not visible to the public network